

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		32103	
Total Threats Mitigated	22	Total Packets Analyzed		3369462442	
High Priority	2	Malicious Packets		3343772212	
Low Priority	20	True Negatives Passed		27077	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		13	
False Positives	35	UDP Flood		2	
FP Rate	0.13%	Uncertain		7	
IF Precision	99.30%				
IF Recall	92.82%	Action		Count	
IF F1-Score	95.95%	Quarantined		5	
IF Accuracy	98.69%	Time Ban		4	
		Blackhole		0	
		Released		4	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.30%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	92.82%	Share of actual attacks successfully flagged
F1-Score	95.95%	Balanced measure combining Precision and Recall
Accuracy	98.69%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.13%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	7.18%	Actual attacks that went undetected
True Positive Rate (TPR)	92.82%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.87%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4991	FN 386
Actual: Normal	FP 35	TN 26691

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	4991	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.30%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	92.82%	Share of actual attacks successfully flagged
F1-Score	95.95%	Balanced measure combining Precision and Recall
Accuracy	98.69%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.13%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	7.18%	Actual attacks that went undetected
True Positive Rate (TPR)	92.82%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.87%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4991	FN 386
Actual: Normal	FP 35	TN 26691

2d. Response Latency

Metric	Value	Description
Detection Time	404.6 ms	Average time taken to detect an attack
Mitigation Response Time	1.0 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.02%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.44%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	4.66%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.18	3	3	2026-09-15 10:12:34	2026-09-15 10:13:55
10.0.0.16	2	2	2026-09-15 10:12:32	2026-09-15 10:13:44
10.0.0.17	2	2	2026-09-15 10:12:32	2026-09-15 10:13:44
10.0.0.19	2	2	2026-09-15 10:12:33	2026-09-15 10:13:45
10.0.0.6	1	1	2026-09-15 10:13:28	2026-09-15 10:14:29
10.0.0.1	1	1	2026-09-15 10:13:28	2026-09-15 10:13:39
10.0.0.5	1	1	2026-09-15 10:13:24	2026-09-15 10:13:35

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 10:12:32	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:12:32	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:12:33	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:12:34	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:12:43	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-15 10:12:43	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-15 10:12:43	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-15 10:12:43	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-15 10:12:44	10.0.0.19	DDoS	SYN Flood	90.0%	Critical	Time Ban (1m)
2026-09-15 10:12:44	10.0.0.18	DDoS	SYN Flood	90.0%	Critical	Time Ban (1m)
2026-09-15 10:12:44	10.0.0.19	DDoS	SYN Flood	90.0%	Critical	Time Ban
2026-09-15 10:12:45	10.0.0.18	DDoS	SYN Flood	90.0%	Critical	Time Ban
2026-09-15 10:13:28	10.0.0.6	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-15 10:13:34	10.0.0.5	DDoS	Uncertain	59.1%	Low	Released
2026-09-15 10:13:39	10.0.0.1	DDoS	Uncertain	58.9%	Low	Released
2026-09-15 10:13:44	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban (2m)
2026-09-15 10:13:44	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban (2m)
2026-09-15 10:13:45	10.0.0.19	DDoS	SYN Flood	93.3%	Critical	Time Ban (2m)
2026-09-15 10:13:45	10.0.0.18	DDoS	SYN Flood	93.7%	Critical	Released
2026-09-15 10:13:55	10.0.0.18	DDoS	SYN Flood	93.7%	Critical	Time Ban (2m)
2026-09-15 10:14:29	10.0.0.6	DDoS	UDP Flood	92.4%	High	Released
2026-09-15 10:14:58	10.0.0.6	DDoS	Uncertain	48.4%	Low	Quarantined

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.6	UDP Flood	0.6018	92.4%	High	Phase 2	1	1m 1s	2026-09-15 10:14:29	Ban Expired
10.0.0.18	SYN Flood	0.6488	93.7%	Critical	Phase 2	3	10s	2026-09-15 10:13:55	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6073	93.7%	Critical	Phase 2	1	1m 11s	2026-09-15 10:13:45	Ban Expired
10.0.0.19	SYN Flood	0.6107	93.3%	Critical	Phase 2	2	1m 12s	2026-09-15 10:13:45	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6327	92.7%	Critical	Phase 2	2	1m 12s	2026-09-15 10:13:44	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6270	92.7%	Critical	Phase 2	2	1m 12s	2026-09-15 10:13:44	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.1	Uncertain	0.5980	58.9%	Low	Phase 1	1	11s	2026-09-15 10:13:39	Attack Stopped
10.0.0.5	Uncertain	0.6044	59.1%	Low	Phase 1	1	11s	2026-09-15 10:13:35	Attack Stopped
10.0.0.18	SYN Flood	0.6436	90.0%	Critical	Phase 2	1	10s	2026-09-15 10:12:44	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6547	90.0%	Critical	Phase 2	1	11s	2026-09-15 10:12:44	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6434	92.7%	Critical	Phase 2	1	11s	2026-09-15 10:12:43	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6363	92.7%	Critical	Phase 2	1	11s	2026-09-15 10:12:43	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE